

Transactions, Concurrency & Recovery - Quick Revision Notes

QUICK REVISION & EXAM CHEAT SHEET

Subject: Database Management Systems | Chapter: Chapter 3: Transaction Processing, Concurrency Control & Recovery | Academic Year: 2023-2024

ACID Properties Breakdown

- * Atomicity -> Recovery Manager (Rollback / Undo Log).
- * Consistency -> Application Logic + DBMS Integrity Constraints.
- * Isolation -> Concurrency Control Manager (Locking / Timestamping).
- * Durability -> Recovery Manager (Write-Ahead Logging / Redo Log).

Isolation Levels & Anomalies

- * Read Uncommitted: Suffers Dirty Read, Non-repeatable Read, Phantom Read.
- * Read Committed: Prevents Dirty Read. Suffers Non-repeatable Read and Phantom Read.
- * Repeatable Read: Prevents Dirty Read and Non-repeatable Read. May suffer Phantom Read.
- * Serializable: Prevents all anomalies. Highest isolation and lowest concurrency.

Locking Protocols & Serializability

- * 2PL guarantees conflict serializability, but can still lead to deadlocks.
- * Strict 2PL releases Exclusive (X) locks ONLY after transaction commit/abort.
- * Precedence Graph: Edge T1 -> T2 if T1 performs conflicting action before T2. Cycle = NOT serializable.